

Stage 2: Practical Build Assessment

1-Page Mini Brief:

Initial State Observations:

- Only one screen exists.
- Datasets are richer/more comprehensive than base product (7 Dataset Files vs 6 columns displayed)
- Only list view, no existing map view or different arrangement.
- Table supports limited filtering/search options:
 - Search by Plate
 - Filter by Status (Dropdown menu)
- No quick inspection or details panel for a record (view vehicle information)
- No sorting capability of rows/records (Order feels arbitrary in terms of importance)
- No pagination despite dataset containing 500 records (page 1, page 2,...)
- No export functionality exists.

The conclusion I draw from these observations is that while fleet managers can technically access the data, the usage is inefficient because important operational information is difficult to identify, link, and inspect quickly.

Problem Statement:

Fleet managers can access the data but the workflow is inefficient because important/needed signals are difficult to identify, inspect and understand quickly.

Target User:

I am targeting the Fleet managers who monitor the ongoing operations and vehicular activity, and investigate the fleet vehicles issues.

Assumptions:

- Fleet managers frequently need to inspect individual vehicle records quickly.
- Operational exceptions are more important than raw data visibility.
- Users benefit from being able to prioritize and sort operationally important records.
- Exporting filtered operational data is a common workflow requirement.
- The assessment prioritizes practical workflow improvements over large infrastructure-heavy features.

Clarifications I Would Seek in a Real Project

- Which workflows consume the exported CSV data and what fields are most important?
- What operational events or exceptions are considered highest priority by fleet managers?
- How frequently is vehicle data refreshed in production?
- What are the expected dataset sizes in real customer environments?
- How is success currently measured for fleet manager workflows and how is in transit status determined?

Proposed Solution:

The proposed solution focuses on improving operational visibility and workflow efficiency through several bounded UI and usability improvements:

- Add a vehicle details panel for quick inspection of individual records.
- Add exception/status highlighting to improve visibility of operationally important vehicles.
- Add export functionality for filtered datasets.
- Improve overall usability while maintaining a lightweight and low-risk implementation.

Scope:

Included: details panel, exception highlighting, export functionality, workflow UX improvements, and automated tests for critical user flows.

Success Metric:

The solution would be considered successful if fleet managers can:

- Identify important vehicles faster
- Inspect records with fewer interactions
- Prioritize operational issues more effectively
- Export relevant operational data efficiently

Tradeoffs:

I prioritized high-impact workflow improvements over larger features such as map visualization. While those features may provide additional value, they would significantly increase implementation complexity, testing requirements, and risk relative to the timeline and impact. The selected improvements provide meaningful operational value while remaining practical, testable, and maintainable.